

SRE CASE STUDY

AZURE ARC HYBRID CLOUD LAB

Parts 1-10: K8s, Azure Arc, Terraform, CI/CD, Ingress TLS, HPA, NetworkPolicy, RBAC, GitOps, Trivy, Defender, cert-manager, KEDA, Helm, OPA/Gatekeeper, Chaos Mesh, Linkerd

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Date: June 2026 (Parts 1-10)

GitHub: github.com/buth11/sre-homelab-azure-arc

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Chapter 1: Context and Architecture

This project demonstrates building a production-grade hybrid environment on GMKtec EVO-X2 (128 GB RAM) connecting Kubernetes with Microsoft Azure through Azure Arc. Three weeks of hands-on practice with documentation of every step and 10 documented incidents.

Layer	Technology	Purpose
Kubernetes	K3s v1.35.4+k3s1	3-node cluster
Ingress	Traefik + TLS	HTTPS domain routing
Autoscaling	HPA (autoscaling/v2)	CPU-based scaling 2-8 replicas
Network Security	NetworkPolicy	Intra-cluster firewall
Access Control	RBAC + ServiceAccount	Least privilege
Cloud Bridge	Azure Arc	Hybrid control plane
Monitoring	Azure Monitor + Prometheus	Observability
IaC	Terraform + Remote State	Infrastructure as Code
CI/CD	Azure DevOps	Pipeline with approval gate

Chapter 2: On-Premise Infrastructure

Node	IP	CPU	RAM
k3s-master (Control Plane)	192.168.122.10	4 vCPU	8 GB
k3s-worker1	192.168.122.11	4 vCPU	16 GB
k3s-worker2	192.168.122.12	4 vCPU	16 GB

Chapter 3: FinOps

The screenshot displays the Microsoft Azure Cost Management dashboard for user Bartosz Suszko. The interface is in Polish. On the left, a navigation pane lists various Azure services and tools. The main area shows the 'Budżety' (Budgets) section. A table lists the budget details for 'SRE-Lab-Budget'.

Nazwa	Zakres	Resetuj okres	Data utworzenia	Data wygaśnięcia	Budżet	Prognostyczne	Oszacowane wy...	Postęp
SRE-Lab-Budget	55df28b8-39ed-58b...	Monthly	1.05.2026	30.04.2028	5,00 USD		\$0.00	0.00%

Screenshot 1: Azure Cost Management — 5 USD/month budget, cost: 0.00 USD.

Chapter 4: Azure Arc

The screenshot displays the Microsoft Azure portal interface for managing an Azure Arc Kubernetes cluster named 'K3s-Homelab'. The left sidebar contains navigation options such as 'Utwórz zasób', 'Strona główna', 'Pulpit nawigacyjny', and 'Wszystkie usługi'. The main content area is divided into several sections:

- Azure Arc | Klastry Kubernetes**: A search bar and a list of resources, including 'K3s-Homelab'.
- K3s-Homelab**: A detailed view of the cluster, showing its status as 'Connected' and its configuration details.
- Właściwości**: A section containing various properties and settings for the cluster.

The cluster details section shows the following information:

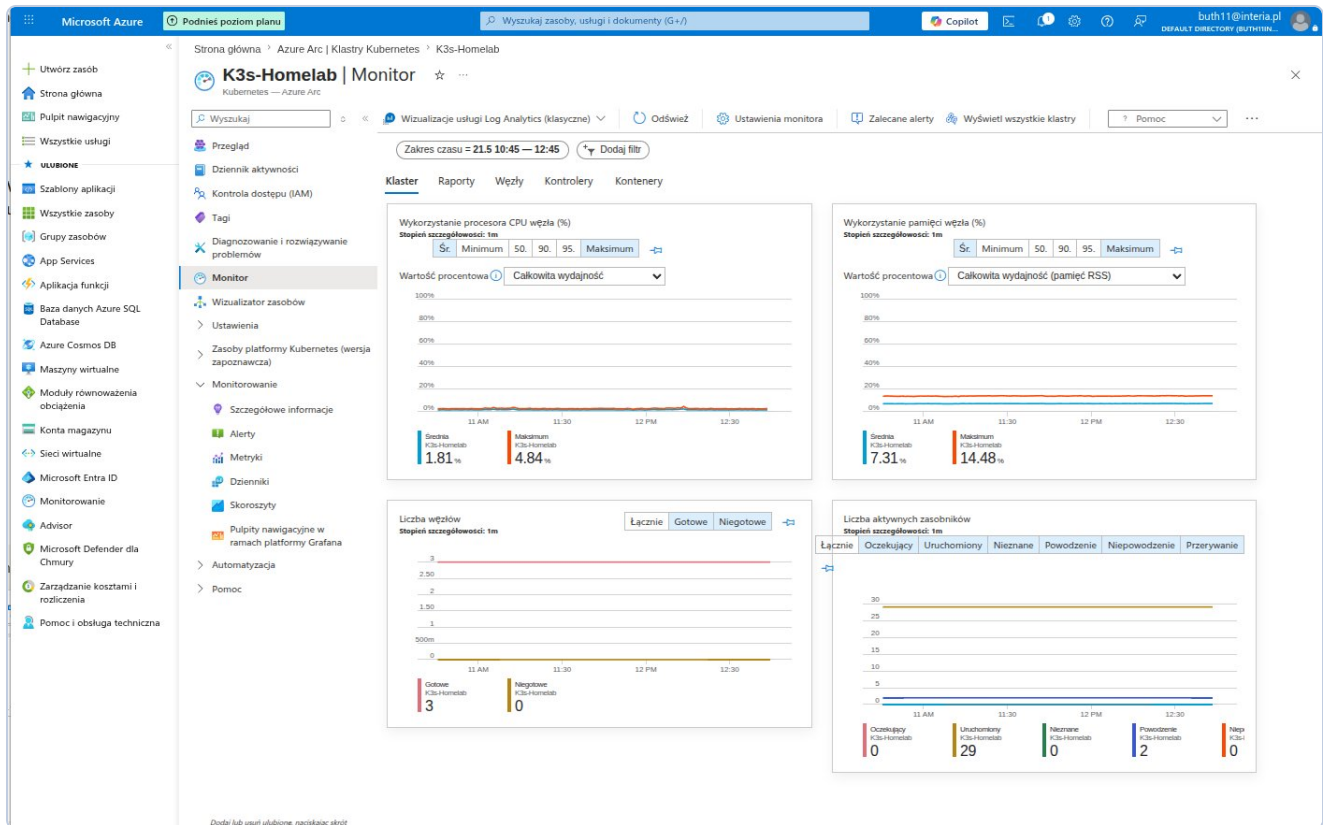
- Podstawowe elementy**:
 - Subskrypcja (przenieś): [Azure subscription 1](#)
 - Identyfikator subskrypcji: b023e8b8-4446-4d4f-9fcf-d20a6c1fa093
 - Grupa zasobów (przenieś): [SRE-Lab-RG](#)
 - Stan: Connected
 - Lokalizacja: West Europe
 - Menedżer floty: [Kliknij tutaj, aby przypisać](#)
- Właściwości**:
 - Kubernetes — Azure Arc**:
 - Wersja agenta: 1.33.0
 - Czas wygaśnięcia certyfikatu tożsamości zarządzanej: August 19, 2026 at 08:20 AM GMT+2
 - Wersja platformy Kubernetes: 1.35.4+k3s1
 - Łączna liczba węzłów: 3
 - Łączna liczba rdzeni: 12
 - Rozszerzenia**: [azuremonitor-containers](#)

Screenshot 2: Azure Arc — K3s-Homelab Connected, v1.35.4+k3s1, 3 nodes, 12 cores.

Chapter 5: Observability

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
{
  "principalId": "a3fe4a8c-7a2e-4b99-8de9-2150d617cb76",
  "tenantId": null,
  "type": "SystemAssigned"
},
"isSystemExtension": false,
"name": "azuremonitor-containers",
"packageUri": null,
"plan": null,
"provisioningState": "Succeeded",
"releaseTrain": "Stable",
"resourceGroup": "SRE-Lab-RG",
"scope": {
  "cluster": {
    "releaseNamespace": "azuremonitor-containers"
  },
  "namespace": null
},
"statuses": [],
"systemData": {
  "createdAt": "2026-05-22T07:44:41.936626+00:00",
  "createdBy": null,
  "createdByType": null,
  "lastModifiedAt": "2026-05-22T07:44:41.936626+00:00",
  "lastModifiedBy": null,
  "lastModifiedByType": null
},
"type": "Microsoft.KubernetesConfiguration/extensions",
"version": null
}
buth11@k3s-master:~$ sudo k3s kubectl get pods -n kube-system | grep ama
[sudo] password for buth11:
ama-logs-9m9x9          3/3      Running    0          37s
ama-logs-9scpr          3/3      Running    0          37s
ama-logs-rs-65b58c7f96-f8lds 2/2      Running    0          37s
ama-logs-spcdp          3/3      Running    0          37s
buth11@k3s-master:~$
```

Screenshot 3: AMA pods Running — 4 agents, 0 restarts.



Screenshot 4: Azure Monitor — CPU avg 1.58%, RAM avg 6.27%, 3 nodes Ready, 29 pods.

Chapter 6: KQL

The screenshot displays the Microsoft Azure portal interface. On the left is the navigation pane with options like 'Create a resource', 'Dashboard', and 'All services'. The main area shows the 'Log Analytics workspace' for 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU'. The 'Logs' section is active, showing a KQL query editor with the following code:

```
1 KubeNodeInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Computer
4 project
5   Computer,
6   Status,
7   KubeletVersion
8 order by Computer asc
```

Below the query editor, the 'Results' tab is selected, displaying a table with 3 rows and 3 columns: Computer, Status, and KubeletVersion.

Computer	Status	KubeletVersion
k3s-master	Ready	v1.35.4+k3s1
k3s-worker1	Ready	v1.35.4+k3s1
k3s-worker2	Ready	v1.35.4+k3s1

The bottom of the interface shows a status bar with '0s 346ms', 'Display time (UTC+00:00)', 'Query details', and '1 - 3 of 3'.

KQL-1: 3 nodes Ready, v1.35.4+k3s1.

Home > Log Analytics workspaces > DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU

Log Analytics workspace

Default Directory (buth11@interna.onmicrosoft.com)

Log Analytics workspace

Search

New Query 1* ... +

Run | Time range: Set in query | Show: 1000 results | KQL mode

```

1 KubePodInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Name, Namespace
4 summarize PodCount = count() by Namespace, PodStatus
5 order by Namespace asc, PodCount desc

```

Namespace	PodStatus	PodCount
azure-arc	Running	12
default	Running	3
kube-system	Running	14
kube-system	Succeeded	2

0s 533ms | Display time (UTC+00:00) | Query details | 1 - 4 of 4

KQL-2: azure-arc 12 Running, default 3 Running, kube-system 14+2.

Home > Log Analytics workspaces > DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU

Log Analytics workspace

Default Directory (buth11@interna.onmicrosoft.com)

Log Analytics workspace

Search

New Query 1* ... +

Run | Time range: Set in query | Show: 1000 results | KQL mode

```

1 KubeEvents
2 where TimeGenerated > ago(24h)
3 project TimeGenerated, Namespace, Name, Reason, Message
4 order by TimeGenerated desc
5 take 20

```

TimeGenerated [UTC]	Namespace	Name	Reason
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-9cpr	FailedMount
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-rs-65b58c7f96-f8ld	FailedMount
5/22/2026, 7:35:32.000 AM	azure-arc	kube-aad-proxy-5f59465457-12...	Unhealthy
5/22/2026, 7:35:25.000 AM	azure-arc	config-agent-7548f59c5-k4zrv	Unhealthy
5/22/2026, 7:35:13.000 AM	k3s-worker2		Rebooted
5/22/2026, 7:35:11.000 AM	k3s-worker1		Rebooted
5/22/2026, 7:35:09.000 AM	kube-system	metrics-server-78d997795-zz...	Unhealthy
5/22/2026, 7:35:07.000 AM	k3s-master		Rebooted
5/21/2026, 10:53:39.000 AM	kube-system	ama-logs-89tpj	Unhealthy
5/21/2026, 10:52:38.000 AM	kube-system	ama-logs-rs-65b58c7f96-nkshw	Unhealthy
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352...	FailedScheduling

0s 287ms | Display time (UTC+00:00) | Query details | 1 - 13 of 13

KQL-3: 13 events 24h — Rebooted, Unhealthy (temporary), FailedScheduling.

The screenshot shows the Microsoft Azure Log Analytics interface. The left sidebar contains navigation options like Home, Dashboard, All services, and Favorites. The main area displays a Log Analytics workspace with a search bar and a list of logs. A KQL query is entered in the 'New Query 1' field:

```
1 KubePodInventory
2 | where TimeGenerated > ago(24h)
3 | summarize arg_max(TimeGenerated, *) by Name, Namespace
4 | where PodRestartCount > 0
5 | project Namespace, Name, PodRestartCount, PodStatus
6 | order by PodRestartCount desc
```

The results table shows 27 pods with restarts, all in a 'Running' state. The table has columns: Namespace, Name, PodRestartCount, and PodStatus.

Namespace	Name	PodRestartCount	PodStatus
azure-arc	extension-manager-85899588c...	3	Running
azure-arc	clusterconnect-agent-5bc68597...	3	Running
kube-system	svclb-traefik-4b75cb6-8b8x...	2	Running
azure-arc	extension-events-collector-6c9...	2	Running
azure-arc	clusteridentityoperator-767768...	2	Running
azure-arc	kube-aad-proxy-5f5465457-42...	2	Running
kube-system	svclb-traefik-4b75cb6-c2gzs...	2	Running
azure-arc	config-agent-75548f59c5-k4zrv...	2	Running
azure-arc	cluster-metadata-operator-6db...	2	Running
kube-system	svclb-traefik-4b75cb6-9587...	2	Running
azure-arc	metrics-agent-6d994764-rc44...	2	Running
azure-arc	controller-manager-7cb948ddff...	2	Running
azure-arc	resource-sync-agent-5f4688cc5...	2	Running
azure-arc	flux-logs-agent-76f8c454c9-9g...	1	Running
azure-arc	logcollector-8bd4468b4-4s2zw...	1	Running
kube-system	ama-logs-nlg5p...	1	Running
kube-system	helm-install-traefik-ggts...	1	Succeeded
kube-system	svclb-aras-demo-app-8581d79...	1	Running
default	aras-demo-app-f78659546-vb...	1	Running
kube-system	coredns-c4dbfbb5f-llkx...	1	Running
kube-system	local-path-provisioner-5c4dc5d...	1	Running
kube-system	metrics-server-786d997795-zz...	1	Running
kube-system	svclb-aras-demo-app-8581d79...	1	Running

KQL-4: 27 pods with restarts after VM reboot — all Running.

The screenshot shows the Microsoft Azure Log Analytics interface. The left sidebar contains navigation options like Home, Dashboard, All services, and Favorites. The main area displays a Log Analytics workspace with a search bar and a list of logs. A KQL query is entered in the 'New Query 1' field:

```
1 KubeNodeInventory
2 | where TimeGenerated > ago(24h)
3 | summarize
4 |   TotalSamples = count(),
5 |   ReadySamples = countif(Status == "Ready")
6 | by Computer
7 | extend AvailabilityPct = round((toReal(ReadySamples) / toReal(TotalSamples)) * 100, 2)
8 | project Computer, AvailabilityPct, ReadySamples, TotalSamples
9 | order by Computer asc
```

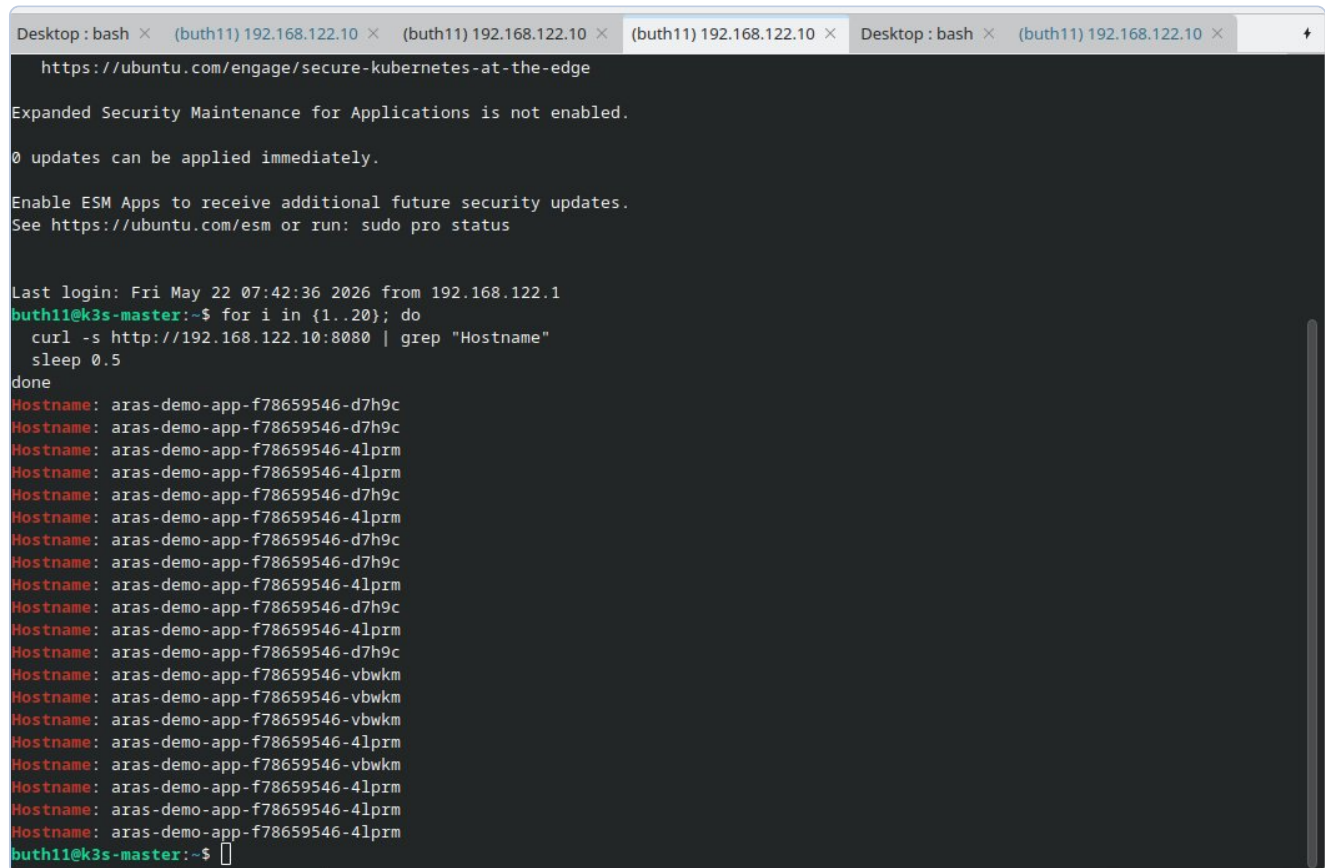
The results table shows 100% availability for all computers. The table has columns: Computer, AvailabilityPct, ReadySamples, and TotalSamples.

Computer	AvailabilityPct	ReadySamples	TotalSamples
k3s-master	100	184	184
k3s-worker1	100	184	184
k3s-worker2	100	184	184

KQL-5: SLO 100% — 184/184 samples Ready.

SLO 100% in 24h window. Error budget intact even after full VM restart.

Chapter 7: Load Balancing



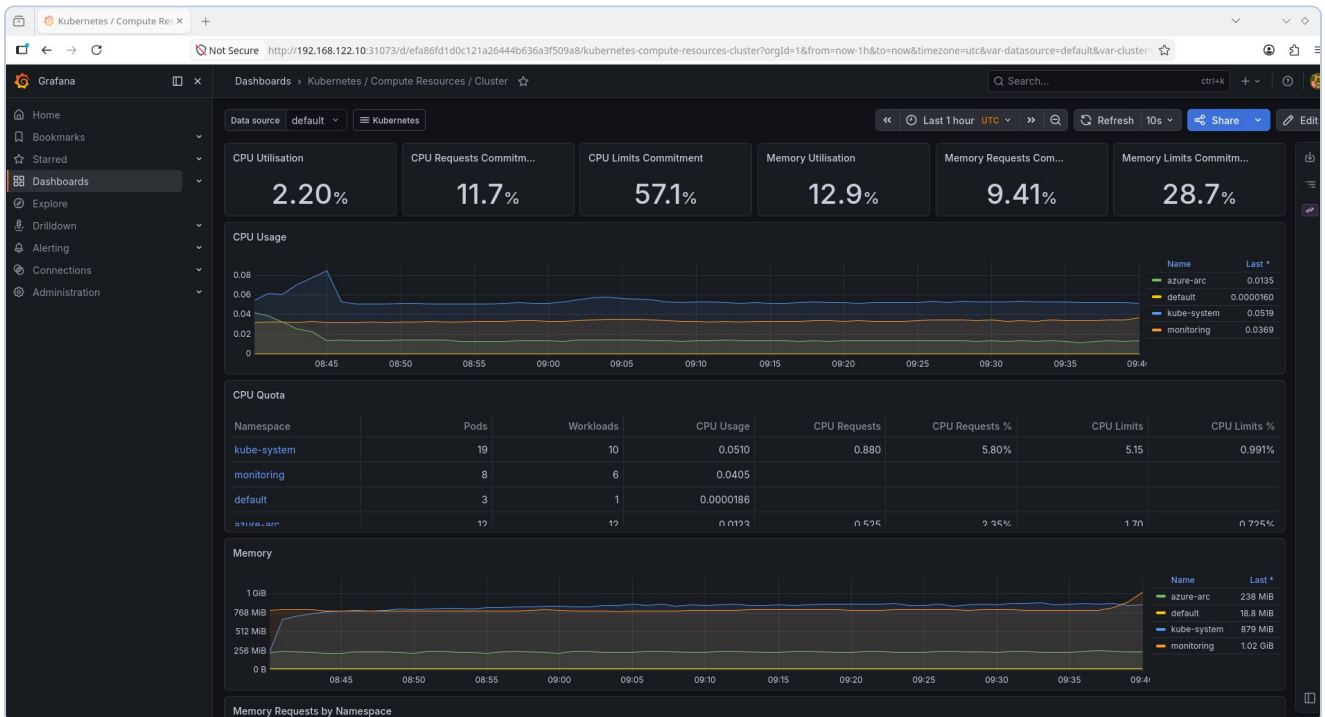
The screenshot shows a terminal window with multiple tabs. The active tab is titled "(buth11) 192.168.122.10". The terminal output includes a security update notification, a login message, and the execution of a loop that sends 20 curl requests to a specific URL. The output of these requests shows a mix of hostnames, indicating a round-robin load balancing distribution across three pods.

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri May 22 07:42:36 2026 from 192.168.122.1
buth11@k3s-master:~$ for i in {1..20}; do
  curl -s http://192.168.122.10:8080 | grep "Hostname"
  sleep 0.5
done
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
buth11@k3s-master:~$
```

Screenshot 5: Round-robin — 20 requests across 3 pods, 900 parallel with zero errors.

Chapter 8: Prometheus & Grafana



Screenshot 6: Grafana — CPU 2.20%, RAM 12.9%, CPU Quota per namespace.

Chapter 9: Terraform IaC

```
Desktop : bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform : bash × (buth11) 192.168.122.10 ×
Plan: 6 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ action_group_id           = (known after apply)
+ budget_name               = "SRE-Lab-Budget"
+ log_analytics_workspace_id = (known after apply)
+ log_analytics_workspace_key = (sensitive value)
+ resource_group_id         = (known after apply)
+ resource_group_name       = "SRE-Lab-RG"
+ subscription_id           = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"

Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
  terraform apply "tfplan"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ terraform apply tfplan
random_string.suffix: Creating...
random_string.suffix: Creation complete after 0s [id=5plwbb]
azurerm_resource_group.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Still creating... [00m10s elapsed]
azurerm_consumption_budget_subscription.sre_lab: Creation complete after 19s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]

Error: A resource with the ID "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG" already exists -
to be managed via Terraform this resource needs to be imported into the State. Please see the resource documentation for "azure
rm_resource_group" for more information.

with azurerm_resource_group.sre_lab,
on main.tf line 26, in resource "azurerm_resource_group" "sre_lab":
26: resource "azurerm_resource_group" "sre_lab" {

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Plan: 6 resources, 0 changes.

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform: bash × (buth11) 192.168.122.10 ×

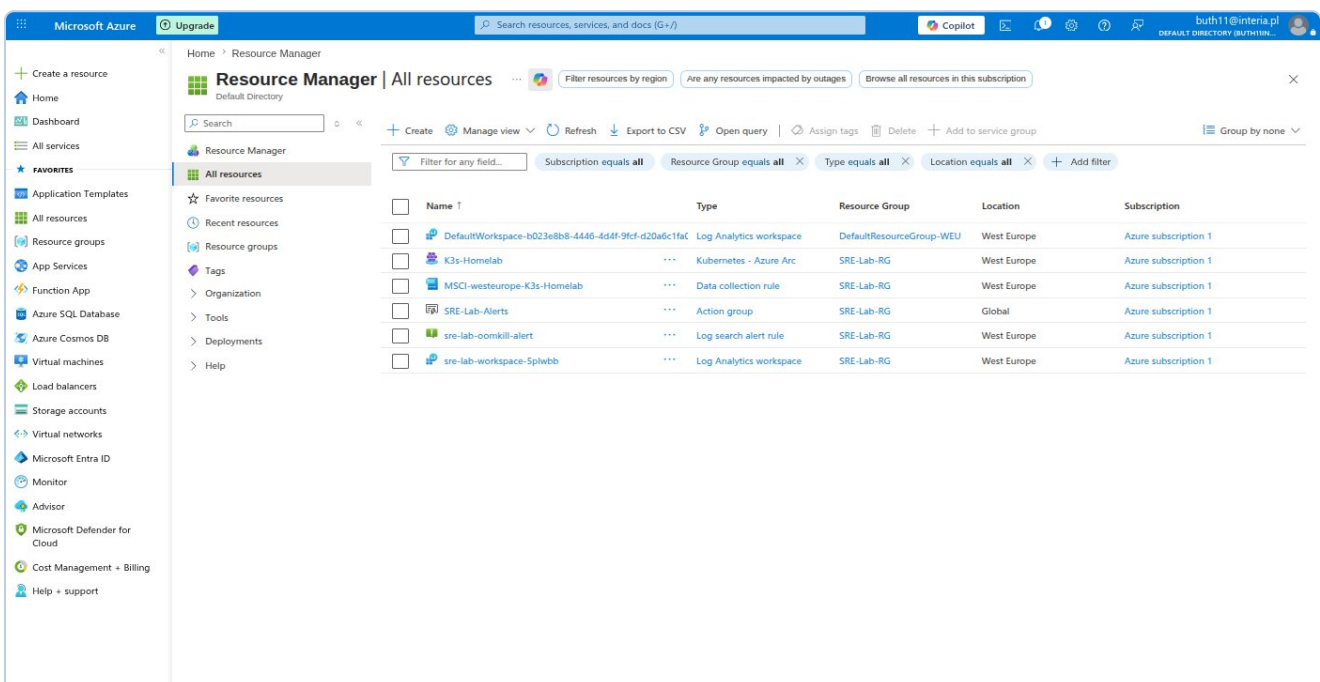
Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
terraform apply "tfplan"
azurerm_resource_group.sre_lab: Modifying... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_resource_group.sre_lab: Modifications complete after 3s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_monitor_action_group.sre_alerts: Creating...
azurerm_log_analytics_workspace.sre_lab: Creating...
azurerm_monitor_action_group.sre_alerts: Creation complete after 4s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m10s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m20s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m30s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m40s elapsed]
azurerm_log_analytics_workspace.sre_lab: Creation complete after 47s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creating...
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creation complete after 5s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

Apply complete! Resources: 3 added, 1 changed, 0 destroyed.

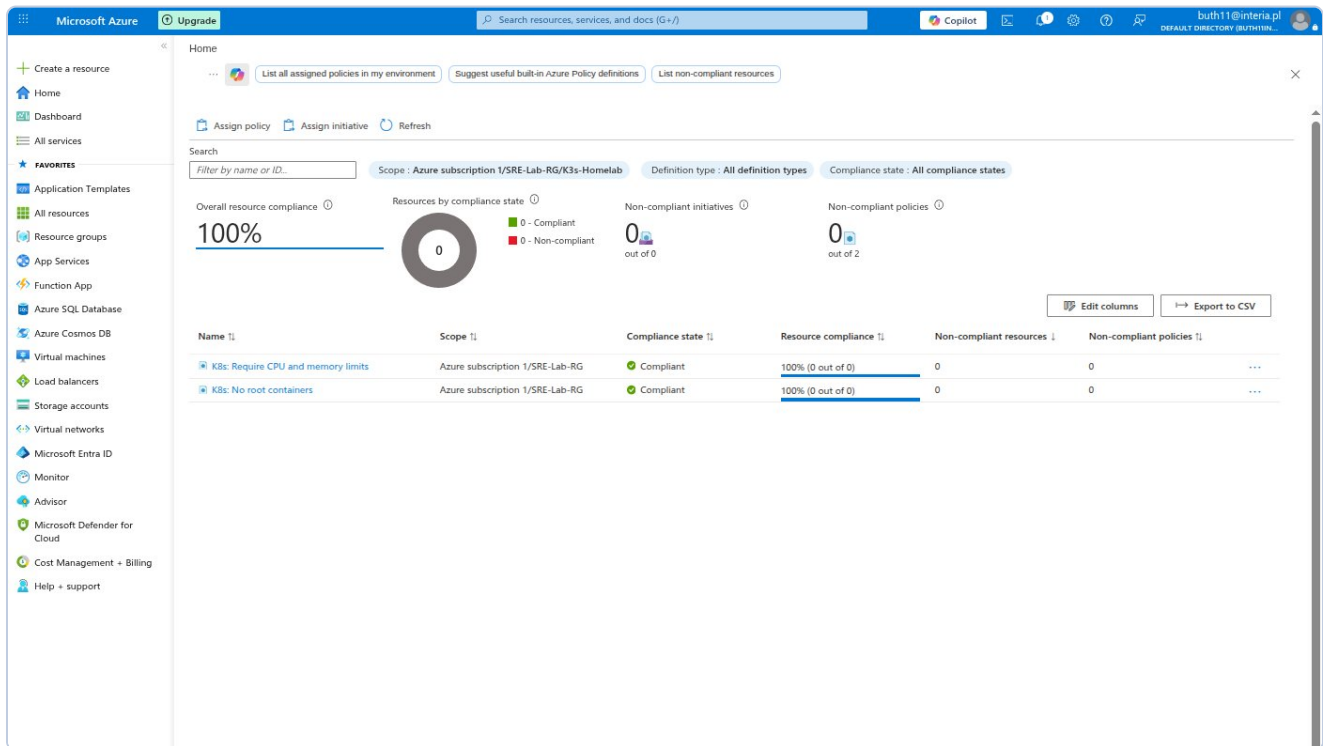
Outputs:
action_group_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts"
budget_name = "SRE-Lab-Budget"
log_analytics_workspace_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb"
log_analytics_workspace_key = <sensitive>
resource_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG"
resource_group_name = "SRE-Lab-RG"
subscription_id = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Apply: 3 added, 1 changed.



Azure Portal: Terraform resources in SRE-Lab-RG.

Chapter 10: Azure Policy



Azure Policy: No root containers + CPU/memory limits — both Compliant.

Chapter 11: GitHub

The screenshot shows the GitHub repository page for 'buth11 / sre-homelab-azure-arc'. The repository is public and has 1 branch (main) and 0 tags. The file list includes: docs, k8s, monitoring, scripts, terraform, .gitignore, and README.md. The README file is selected, showing the title 'SRE Homelab: Azure Arc Hybrid Cloud Lab' and a description: 'A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.' The right sidebar shows the repository's statistics: 0 stars, 0 watching, and 0 forks. The 'Languages' section shows a bar chart with Shell at 58.4% and HCL at 41.6%.

Repository: **sre-homelab-azure-arc** (Public)

Initial commit: K3s + Azure Arc SRE Homelab · 3d8283c · 8 minutes ago · 1 Commit

File	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
docs	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
k8s	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
monitoring	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
scripts	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
terraform	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
.gitignore	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
README.md	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago

README

SRE Homelab: Azure Arc Hybrid Cloud Lab

A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.

Overview

About

No description, website, or topics provided.

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 1

Languages

Shell 58.4% HCL 41.6%

GitHub repo sre-homelab-azure-arc — public, Shell 58.4%, HCL 41.6%.

Chapter 12: Part 2 — Service Principal, Remote State, CI/CD

Remote State — 14.53 KiB, encrypted, versioning. State lock on every plan.

DevOps Pipeline: validate + plan (37s) → Approval Gate → apply (49s) = 4m 24s total.

Microsoft Azure Upgrade Search resources, services, and docs (G+/I) buth11@interia.pl DEFAULT DIRECTORY (BUTH11IN...

Create a resource

Home Dashboard All services FAVORITES Application Templates All resources Resource groups App Services Function App Azure SQL Database Azure Cosmos DB Virtual machines Load balancers Storage accounts Virtual networks Microsoft Entra ID Monitor Advisor Microsoft Defender for Cloud Cost Management + Billing Help + support

sre-homelab.tfstate

Blob

Save Discard Download Refresh Delete Change tier Acquire lease Break lease Give feedback

Overview Versions Snapshots Edit Generate SAS

Properties

URL	https://sreftfstate5496.blob...
LAST MODIFIED	5/23/2026, 8:23:07 PM
CREATION TIME	5/23/2026, 8:23:07 PM
VERSION ID	2026-05-23T18:23:07.9861361Z
TYPE	Block blob
SIZE	14.53 KiB
ACCESS TIER	Hot (Inferred)
ACCESS TIER LAST MODIFIED	N/A
ARCHIVE STATUS	-
REHYDRATE PRIORITY	-
SERVER ENCRYPTED	true
ETAG	0x8DEB8F856BD8961
VERSION-LEVEL IMMUTABILITY POLICY	Disabled
CACHE-CONTROL	
CONTENT-TYPE	application/json
CONTENT-MD5	2mHfbTY53lmCzqo/RW5+MQ...
CONTENT-ENCODING	
CONTENT-LANGUAGE	
CONTENT-DISPOSITION	
LEASE STATUS	Unlocked
LEASE STATE	Available
LEASE DURATION	-
COPY STATUS	-
COPY COMPLETION TIME	-

Undelete

Metadata

Key	Value

Blob index tags

Key	Value

Screenshot W2-2: Azure Portal — blob properties: 14.53 KiB, encrypted, versioning active.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # terraform plan pobierze state z Azure Blob zamiast lokalnego pliku
# Powinno pokazać "No changes" bo infrastruktura jest aktualna
terraform plan
Acquiring state lock. This may take a few moments...
random_string.suffix: Refreshing state... [id=5plwbb]
data.azurearm_subscription.current: Reading...
azurearm_resource_group.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurearm_monitor_action_group.sre_alerts: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurearm_log_analytics_workspace.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
data.azurearm_subscription.current: Read complete after 1s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093]
azurearm_consumption_budget_subscription.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]
azurearm_monitor_scheduled_query_rules_alert_v2.oom_kill: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
Releasing state lock. This may take a few moments...
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Screenshot W2-3: terraform plan — Acquiring/Releasing state lock. No changes.

The screenshot shows the Azure DevOps web interface. On the left is a sidebar with navigation options: Overview, Boards, Repos, Pipelines, Environments, Releases, Library (selected), Task groups, Deployment groups, Test Plans, and Artifacts. The main area is titled 'Library > terraform-credentials'. Below this, there are tabs for 'Variable group', 'Save', 'Clone', 'Security', 'Pipeline permissions', 'Approvals and checks', and 'Help'. The 'Properties' section shows the 'Variable group name' as 'terraform-credentials' and a 'Description' field. There is a toggle switch for 'Link secrets from an Azure key vault as variables' which is currently off. The 'Variables' section contains a table with the following data:

Name ↑	Value	Icon
ARM_CLIENT_ID	54c7e017-a745-4e1e-b9da-d74f4901a3de	
ARM_CLIENT_SECRET	*****	🔒
ARM_SUBSCRIPTION_ID	b023e8b8-4446-4d4f-9fcf-d20a6c1fa093	
ARM_TENANT_ID	fd7288c4-e003-40fd-b3cf-9c0d1821cf5d	

At the bottom of the variables list is a '+ Add' button. The bottom of the sidebar shows 'Project settings' with a double arrow icon.

Screenshot W2-4: Azure DevOps Variable Group — ARM_* with hidden CLIENT_SECRET.

Azure DevOps

buth11

/ sre-homelab-arc

/ Pipelines

/ Environments

/ production

Search

BS

sre-homelab-arc

+

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Releases

Library

Task groups

Deployment groups

Test Plans

Artifacts

Project settings

<<

production

Automatically created environment

Add resource

Deployments

Approvals and checks

Check execution order

Checks will run by order, which is pre-determined by check types. Checks within the same order will run in parallel.
[Learn more](#)

+ Add new

Order	Display name	Type	Timeout	Details
1	All approvers must approve	Approvals	30d	8%

Screenshot W2-5: Environment production — Approval Gate active.

Bartosz Suszko | SRE Case Study | 21 / 33

Azure DevOps | sre-homelab-arc | Pipelines | buth11.sre-homelab-azure-... | 20260523.1

Overview
Boards
Repos
Pipelines
Pipelines
Environments
Releases
Library
Task groups
Deployment groups
Test Plans
Artifacts

Project settings

Jobs in run #20260523.1

buth11.sre-homelab-azure-arc

Terraform Validate & Plan

terraform plan	34s
Initialize job	<1s
Checkout buth1...	2s
Install Terraform...	2s
terraform init	7s
terraform validate	1s
terraform plan	15s
Publish terrafor...	3s
Post-job: Chec...	<1s
Finalize Job	<1s

Terraform Apply

- terraform apply

terraform plan

View raw log

```
1 Agent: Azure Pipelines 1
2 Started: Just now
3 Duration: 34s
4
5 Job preparation parameters
44 1 artifact produced
45 Job live console data:
46 Starting: terraform plan
47 Async Command Start: DetectDockerContainer
48 Async Command End: DetectDockerContainer
49 Async Command Start: DetectDockerContainer
50 Async Command End: DetectDockerContainer
51 Finishing: terraform plan
```

Screenshot W2-6: Stage 1 — all steps green, 1 artifact.

Azure DevOps | sre-homelab-arc | Pipelines | #20260523.1 • feat: add Azure DevOps CI/CD pipeline for Terraform

Run new

This run is being retained as one of 3 recent runs by pipeline. View retention leases

Summary Environments Associated pipelines Code Coverage

Manually run by Bartosz Suszko
Today at 9:13 PM 4m 24s
View change Parameters

Repository and version: sre-homelab-arc/main e9da4bfb
Related: 0 work items, 1 published
Tests and coverage: Get started

Stages Jobs

Terraform Validate ... 1 job completed 37s 1 artifact

Terraform Apply 1 job completed 49s 1 checks passed

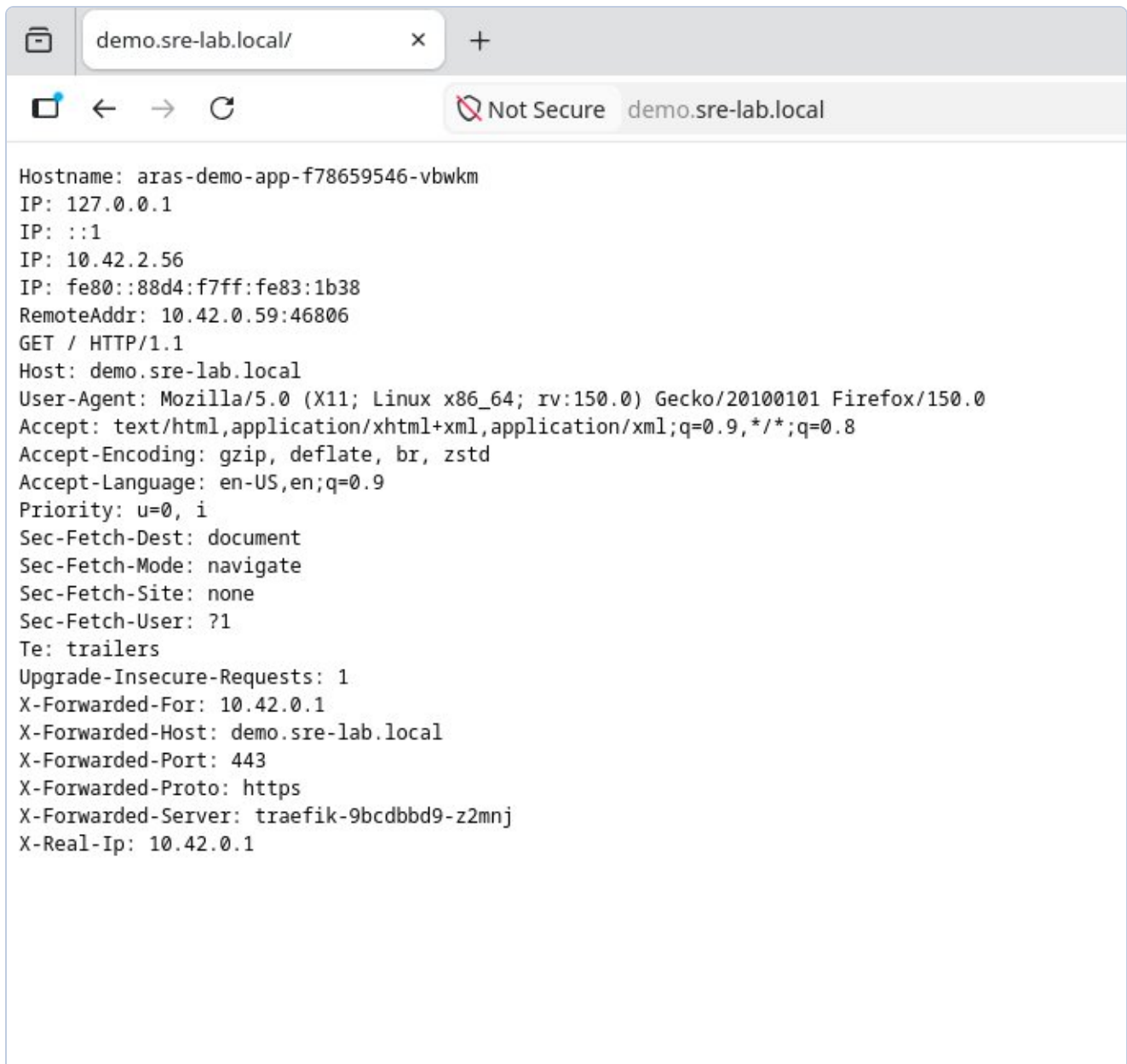
Project settings

Screenshot W2-7: Pipeline complete — both stages green, 4m 24s.

Chapter 13: Part 3 — Advanced Kubernetes Operations

PART 3: Ingress TLS, HPA, NetworkPolicy, RBAC

Traefik HTTPS, self-signed TLS, HPA 2-8 replicas (860% CPU), NetworkPolicy deny-all, RBAC least privilege.



Screenshot W3-1: Firefox browser — <https://demo.sre-lab.local> working. Visible: Pod hostname, X-Forwarded-Proto: https, X-Forwarded-Server: traefik-9bcdbbd9-z2mnj, X-Forwarded-Port: 443.

HPA — CPU-based pod autoscaling

Horizontal Pod Autoscaler configured for the aras-demo-app Deployment. HPA queries metrics-server every 15 seconds for CPU usage. When average pod CPU exceeds 50% — it adds replicas. When it drops below 50% for 5 minutes — it removes pods.

- Range: minReplicas: 2, maxReplicas: 8, targetCPU: 50%
- Load test: `ab -n 50000 -c 50` — 1621 req/s over 30 seconds
- Scale-up: CPU 860% → automatic scaling to 8 replicas
- Scale-down: CPU 6% after 5 minutes → return to 2 replicas (minimum)

```
Every 5.0s: sudo k3s kubectl get hpa &... k3s-master: Sun May 24 09:32:49 2026
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS
aras-demo-hpa	Deployment/aras-demo-app	cpu: 860%/50%	2	8
REPLICAS	AGE			
4	9m55s			

aras-demo-app-5b58b48d6d-5vw14	1/1	Running 0	24s	
aras-demo-app-5b58b48d6d-7mbln	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-859c7	1/1	Running 0	6m32s	
aras-demo-app-5b58b48d6d-96qs4	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-phz5d	1/1	Running 0	6m27s	
aras-demo-app-5b58b48d6d-sbv4q	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-xx94p	1/1	Running 0	24s	
aras-demo-app-5b58b48d6d-zrmgs	1/1	Running 0	10s	

Screenshot W3-2: HPA scale-up — cpu: 860%/50%, REPLICAS: 8 (max), 8 pods Running. Automatic scaling in action during ab load test.

```
Every 5.0s: sudo k3s kubectl get hpa &... k3s-master: Sun May 24 09:38:17 2026

NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  RE
PLICAS    AGE
aras-demo-hpa  Deployment/aras-demo-app  cpu: 6%/50%    2         8         8
15m
---
aras-demo-app-5b58b48d6d-phz5d  1/1    Running    0         11m
aras-demo-app-5b58b48d6d-zrmgs  1/1    Running    0         5m38s
```

Screenshot W3-3: HPA scale-down — cpu: 6%/50%, REPLICAS: 2 (min). After 5 minutes without traffic HPA automatically removed 6 unnecessary pods.

NetworkPolicy — intra-cluster firewall

By default all pods in Kubernetes can communicate with all pods. NetworkPolicy defines rules like a firewall — which namespaces and pods can connect to the demo application. Deployed a policy blocking all incoming traffic except from allowed sources.

- Allowed: kube-system (Traefik), monitoring (Prometheus), aras-demo-app replicas
- Blocked: everything else
- Verification: unauthorized pod — timeout/blocked. Traefik — application response

RBAC — least privilege for applications

Role-Based Access Control — every application should have its own account (ServiceAccount) with exactly the permissions it needs. Least privilege principle: if an application doesn't need to delete pods, it shouldn't be able to — even if compromised by an attacker.

- ServiceAccount: aras-demo-sa (instead of shared default)
- Role: get/list/watch on Pods and ConfigMaps — nothing more
- Verification: can-i list pods = yes, can-i delete pods = no

```
buth11@k3s-master:~$ # Test 1: nieautoryzowany Pod probuje polaczyc sie z aplik
acja
# timeout 5 = czekamy max 5 sekund na odpowiedz
# Oczekiwany wynik: brak odpowiedzi / timeout - polityka blokuje
echo "=== Test: nieautoryzowany dostep (powinien byc zablokowany) ==="
sudo k3s kubectl run test-blocked \
  --image=curlimages/curl \
  --rm -it \
  --restart=Never \
  -- curl -s --max-time 5 http://aras-demo-app:8080 || echo "ZABLOKOWANY - Netw
orkPolicy dziala"

# Test 2: sprawdzamy ze Ingress (Traefik) nadal dziala
# Traefik jest w kube-system wiec ma dostep
echo "=== Test: dostep przez Ingress (powinien dzialac) ==="
curl -sk https://demo.sre-lab.local | head -2
=== Test: nieautoryzowany dostep (powinien byc zablokowany) ===
pod "test-blocked" deleted from default namespace
pod default/test-blocked terminated (Error)
ZABLOKOWANY - NetworkPolicy dziala
=== Test: dostep przez Ingress (powinien dzialac) ===
buth11@k3s-master:~$ # Test dostepu przez Traefik - uzywamy IP zamiast domeny
# Traefik nasluchuje na porcie 80 i 443 na wszystkich wezlach
# -H podajemy naglowek Host zeby Traefik wiedzial do ktorego serwisu kierowac
echo "=== Test dostep przez Traefik (kube-system) ==="
curl -s -H "Host: demo.sre-lab.local" http://192.168.122.10 | head -3

# Dodatkowo sprawdzamy ze bezposredni dostep do portu 8080
# przez serwis nadal dziala (LoadBalancer Service jest poza NetworkPolicy)
echo "=== Test dostep przez LoadBalancer Service ==="
curl -s http://192.168.122.10:8080 | head -3
=== Test dostep przez Traefik (kube-system) ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
=== Test dostep przez LoadBalancer Service ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
buth11@k3s-master:~$
```

Screenshot W3-4: RBAC verification — ServiceAccount aras-demo-sa: list pods=yes (allowed), delete pods=no (blocked). Least privilege principle confirmed.

Chapter 14: Postmortem POST-001

Field	Value
Severity	P2
Duration	Under 7 minutes
Root Cause	Typo: ks3-worker1 instead of k3s-worker1
Status	Resolved — regex validation in scripts

Chapter 15: Results and Conclusions (Parts 1-3)

Metric	Part 1	Part 2	Part 3
Nodes	3/3 Ready	—	—
SLO	100%	—	—
Cost	\$0.00	\$0.00	\$0.00
Terraform state	Local	Remote (Blob)	—
CI/CD	—	DevOps 4m24s	—
Ingress TLS	—	—	HTTPS verified
HPA	—	—	3 to 8 to 2
NetworkPolicy	—	—	Blocked
RBAC	—	—	Least privilege
Issues	6	8	10

Chapter 16: Troubleshooting Log

ISSUE-001 KQL: CPUCapacityNanoCores column missing

project operator: Failed to resolve scalar expression named CPUCapacityNanoCores

Resolution: KubeNodeInventory | getschema — simplified query to available columns.

ISSUE-002 Azure Policy: MissingPolicyParameter cpuLimit/memoryLimit

(MissingPolicyParameter) policy assignment k8s-resource-limits is missing cpuLimit, memoryLimit

Resolution: Added --params with cpuLimit: 500m and memoryLimit: 512Mi.

ISSUE-003 Terraform: Resource Group already exists (brownfield)

Error: A resource with the ID .../SRE-Lab-RG already exists

Resolution: terraform import azurearm_resource_group.sre_lab /subscriptions/.../SRE-Lab-RG

ISSUE-004 Terraform: Saved plan is stale after import

Error: Saved plan is stale – state was changed after plan was created

Resolution: New terraform plan -out=tfplan after import.

ISSUE-005 Container Insights missing after VM restart

Resolution: az k8s-extension create --name azuremonitor-containers — agents returned in 37s.

ISSUE-006 POST-001: etcd Split-Brain (ks3 vs k3s)

FATA[0034] Node password rejected, duplicate hostname

Resolution: kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent. Scripts updated with regex validation.

ISSUE-007 Azure DevOps: TerraformInstaller task missing

A task is missing: TerraformInstaller

Resolution: Install from Marketplace: ms-devlabs.custom-terraform-tasks.

ISSUE-008 Pipeline: permission for Variable Group

This pipeline needs permission to access a resource

Resolution: View → Permit — one-time authorization.

ISSUE-009 sudo does not expand ~ in paths

error: Cannot read file ~/ingress/server.crt, open ~/ingress/server.crt: no such file or directory

Root Cause: sudo runs the command as root which has a different home directory. Tilde ~ expands to /root/ instead of /home/buth11/.

Resolution: Use full path: /home/buth11/ingress/server.crt instead of ~/ingress/server.crt

ISSUE-010 HPA shows cpu: unknown — missing resource requests

```
NAME TARGETS MINPODS MAXPODS REPLICAS — cpu: <unknown>/50%
```

Root Cause: HPA calculates CPU percentage as $\text{current_usage}/\text{requests} \times 100$. Without defined requests in the Deployment, HPA has no baseline to calculate percentages from.

Resolution: Added resources.requests to Deployment: cpu: 10m, memory: 32Mi. After rolling update HPA started showing correct values: cpu: 0%/50%.

Chapter 17: Part 4 — GitOps with FluxCD 2.8.8

FluxCD bootstrapped from GitHub repository via SSH key. Four controllers installed in flux-system namespace — all healthy. Kustomization gitops-demo synchronizes the gitops-demo namespace.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ # Part 4
flux get all && kubectl get pods -n flux-system && kubectl get pods -n mdc

# Part 5
kubectl get clusterissuer && kubectl get certificate -n default && curl -sk https://demo.sre-lab.local | grep -E "Hostname|X-Forwarded"
NAME                                REVISION                SUSPENDED    READY    MESSAGE
gitrepository/flux-system           main@sha1:2658d5e5      False       True     stored artifact for revision 'main@sha1:2658d5e5'

NAME                                REVISION                SUSPENDED    READY    MESSAGE
kustomization/flux-system           main@sha1:2658d5e5      False       True     Applied revision: main@sha1:2658d5e5
kustomization/gitops-demo           main@sha1:2658d5e5      False       True     Applied revision: main@sha1:2658d5e5

NAME                                READY    STATUS    RESTARTS    AGE
helm-controller-5984cc88cf-94jkw    1/1      Running   1 (121m ago) 2d3h
kustomize-controller-d6bb746df-9khwf 1/1      Running   1 (121m ago) 2d3h
notification-controller-7bc5c97f8d-jdf1b 1/1      Running   1 (121m ago) 2d3h
source-controller-745c67ff9-2g8qd    1/1      Running   1 (121m ago) 2d3h
NAME                                READY    STATUS    RESTARTS    AGE
microsoft-defender-collectors-ds-cn56h 2/2      Running   2 (121m ago) 24h
microsoft-defender-collectors-ds-pz7cf 2/2      Running   2 (121m ago) 24h
microsoft-defender-collectors-ds-tfq8b 2/2      Running   2 (121m ago) 24h
microsoft-defender-pod-collector-misc-5d785599b4-zr2xw 1/1      Running   1 (121m ago) 24h
microsoft-defender-publisher-ds-5d6cs 1/1      Running   1 (121m ago) 24h
microsoft-defender-publisher-ds-7qgjw 1/1      Running   1 (121m ago) 24h
microsoft-defender-publisher-ds-hkq5m 1/1      Running   1 (121m ago) 24h
NAME                                READY    AGE
selfsigned-issuer                   True     119m
sre-lab-ca-issuer                   True     119m
NAME                                READY    SECRET    AGE
demo-sre-lab-tls                   True     demo-sre-lab-tls 118m
Hostname: aras-demo-app-5b58b48d6d-spwkm
X-Forwarded-For: 10.42.0.1
X-Forwarded-Host: demo.sre-lab.local
X-Forwarded-Port: 443
X-Forwarded-Proto: https
X-Forwarded-Server: traefik-9bcd9bd9-z2mnj
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$
```

Screenshot W4-1: flux get all — gitrepository/flux-system and both kustomizations Ready: True. kubectl get pods -n flux-system — 4 controllers Running. GitOps test: changed replicas 2→3 via git push, third replica appeared automatically.

Chapter 18: Part 4 — Trivy — Container Image CVE Scanning

Trivy 0.70.0 scanned traefik/whoami:latest. All vulnerabilities are in stdlib Go v1.24.1.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ trivy image traefik/whoami:latest --severity CRITICAL,HIGH --format table 2>/dev/null | grep -A3 "CRITICAL\|Report Summary"
Report Summary
+-----+-----+-----+-----+-----+-----+-----+
| Target | Type | Vulnerabilities | Secrets |
+-----+-----+-----+-----+-----+-----+-----+
--
Total: 30 (HIGH: 29, CRITICAL: 1)

+-----+-----+-----+-----+-----+-----+-----+
| Library | Vulnerability | Severity | Status | Installed Version | Fixed Version | Title |
+-----+-----+-----+-----+-----+-----+-----+
| stdlib | CVE-2025-68121 | CRITICAL | fixed | v1.24.1 | 1.24.13, 1.25.7, 1.26.0-rc.3 | crypto/tls: crypto/tls: Incorrect certificate validation during TLS session resumption https://avd.aquasec.com/nvd/cve-2025-68121 |
+-----+-----+-----+-----+-----+-----+-----+
```

Screenshot W4-2: Trivy Report Summary — Total: 30 (HIGH: 29, CRITICAL: 1). CVE-2025-68121 CRITICAL — crypto/tls: Incorrect certificate validation during TLS session resumption. Fixed: Go 1.24.13, 1.25.7.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ trivy image traefik/whoami:latest --severity CRITICAL,HIGH --format table 2>/dev/null | tail -20
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-39823 | | | | | CVE-2026-27142 fixed a vulnerability in which URLs were not correctly ..... https://avd.aquasec.com/nvd/cve-2026-39823 |
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-39825 | | | | | ReverseProxy can forward queries containing parameters not visible to ... https://avd.aquasec.com/nvd/cve-2026-39825 |
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-39826 | | | | | If a trusted template author were to write a <script> tag containing... https://avd.aquasec.com/nvd/cve-2026-39826 |
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-39836 | | | | | Panic in Dial and LookupPort when handling NUL byte on Windows in... https://avd.aquasec.com/nvd/cve-2026-39836 |
+-----+-----+-----+-----+-----+-----+-----+
| CVE-2026-42499 | | | | | Pathological inputs could cause DoS through consumePhrase when parsing ... https://avd.aquasec.com/nvd/cve-2026-42499 |
+-----+-----+-----+-----+-----+-----+-----+
```

Screenshot W4-3: Trivy — HIGH CVEs list (excerpt). CVE-2026-39823, CVE-2026-39825 (ReverseProxy), CVE-2026-39826 (XSS), CVE-2026-39836 (NUL panic), CVE-2026-42499 (DoS).

Chapter 19: Part 4 — Defender for Cloud

Defender for Containers enabled at Standard tier (30-day trial). 7 DaemonSet pods in mdc namespace — one per node.

Screenshot W4-1 (above, `kubectl get pods -n mdc` section) shows results: `microsoft-defender-collectors-ds-*` (3x, 2/2 Running), `microsoft-defender-publisher-ds-*` (3x, 1/1 Running), `microsoft-defender-pod-collector-misc` (1/1 Running).

Chapter 20: Part 5 — cert-manager v1.20.2

cert-manager installed via Helm. Automatic TLS certificate issuance and renewal. Certificate demo-sre-lab-tls issued in 22 seconds from Ingress annotation alone.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ # Part 4
flux get all && kubectl get pods -n flux-system && kubectl get pods -n mdc

# Part 5
kubectl get clusterissuer && kubectl get certificate -n default && curl -sk https://demo.sre-lab.local | grep -E "Hostname|X-Forwarded"
NAME                                REVISION    SUSPENDED    READY    MESSAGE
gitrepository/flux-system           main@sha1:2658d5e5    False        True     stored artifact for revision 'main@sha1:2658d5e5'

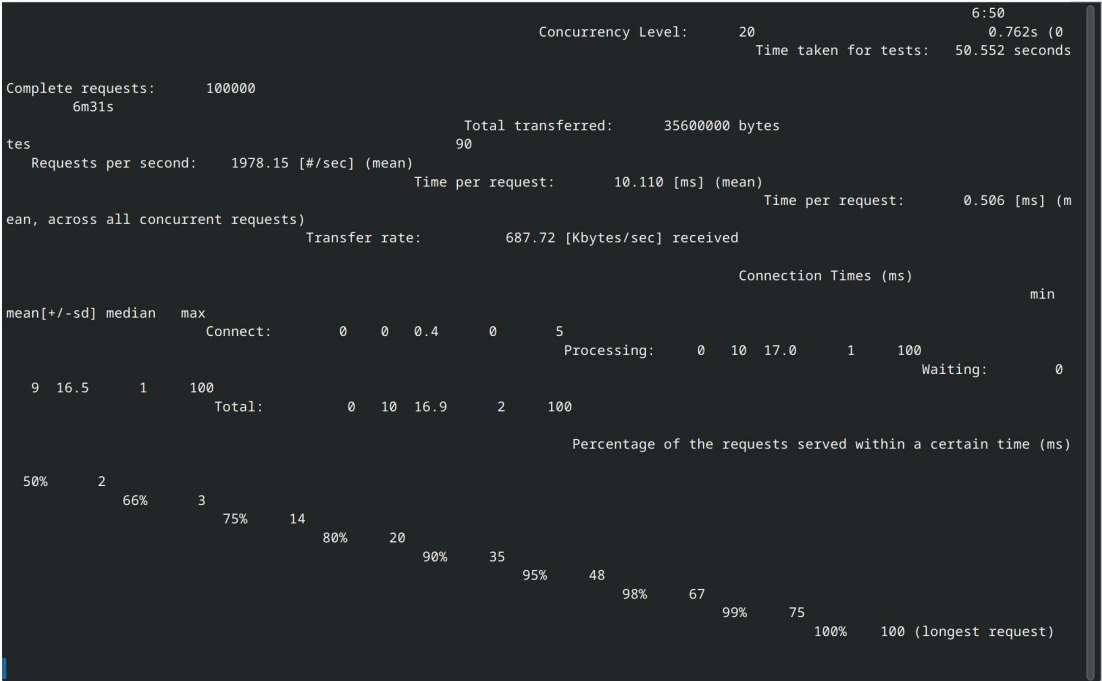
NAME                                REVISION    SUSPENDED    READY    MESSAGE
kustomization/flux-system           main@sha1:2658d5e5    False        True     Applied revision: main@sha1:2658d5e5
kustomization/gitops-demo           main@sha1:2658d5e5    False        True     Applied revision: main@sha1:2658d5e5

NAME                                READY    STATUS    RESTARTS    AGE
helm-controller-5984cc88cf-94jkw    1/1      Running   1 (121m ago)  2d3h
kustomize-controller-d6bb746df-9khwf 1/1      Running   1 (121m ago)  2d3h
notification-controller-7bc5c97f8d-jdf1b 1/1      Running   1 (121m ago)  2d3h
source-controller-745c67ff9-2g8qd    1/1      Running   1 (121m ago)  2d3h
NAME                                READY    STATUS    RESTARTS    AGE
microsoft-defender-collectors-ds-cn56h 2/2      Running   2 (121m ago)  24h
microsoft-defender-collectors-ds-pz7cf 2/2      Running   2 (121m ago)  24h
microsoft-defender-collectors-ds-tfq8b 2/2      Running   2 (121m ago)  24h
microsoft-defender-pod-collector-misc-5d785599b4-zr2xw 1/1      Running   1 (121m ago)  24h
microsoft-defender-publisher-ds-5d6cs 1/1      Running   1 (121m ago)  24h
microsoft-defender-publisher-ds-7qgjw 1/1      Running   1 (121m ago)  24h
microsoft-defender-publisher-ds-hkq5m 1/1      Running   1 (121m ago)  24h
NAME                                READY    AGE
selfsigned-issuer                    True     119m
sre-lab-ca-issuer                     True     119m
NAME                                READY    SECRET    AGE
demo-sre-lab-tls                     True     demo-sre-lab-tls 118m
Hostname: aras-demo-app-5b58b48d6d-spwkm
X-Forwarded-For: 10.42.0.1
X-Forwarded-Host: demo.sre-lab.local
X-Forwarded-Port: 443
X-Forwarded-Proto: https
X-Forwarded-Server: traefik-9bcd9bd9-z2mnj
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$
```

Screenshot W5-1: kubectl get clusterissuer — selfsigned-issuer True, sre-lab-ca-issuer True. kubectl get certificate -n default — demo-sre-lab-tls Ready: True, AGE: 118m. curl verification: X-Forwarded-Proto: https, X-Forwarded-Port: 443, X-Forwarded-Server: traefik.

Chapter 21: Part 6 — KEDA Event-Driven Autoscaling

KEDA scales applications based on Prometheus metrics (container_cpu_usage_seconds_total). ScaledObject: minReplicaCount=1, maxReplicaCount=8, cooldownPeriod=30s.



Screenshot W6-1: ab baseline — 100000 requests, 1978 req/s, 50.552s.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:52:21 PM CEST in 0.575s (0)
```

NAME	SCALETARGETKIND	SCALETARGETNAME	MIN	MAX	READY	ACTIVE	FALLBACK	PAUSED	TRIGGERS	AUTHENTICATIONS
aras-demo-keda	apps/v1.Deployment	aras-demo-app	1	8	True	True	False	False	prometheus	2

aras-demo-app-5b58b48d6d-spwkm	1/1	Running	0	7m46s						

Screenshot W6-2: KEDA ACTIVE: True — scale-up triggered by Prometheus CPU metric. 2 replicas Running.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:53:04 PM CEST
in 0.998s (0)
NAME          SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  A
GE
aras-demo-keda  apps/v1.Deployment  aras-demo-app    1    8    True   False   False     False   prometheus                2
2m
---
aras-demo-app-5b58b48d6d-dhfhb  1/1    Running  0      13s
aras-demo-app-5b58b48d6d-q9qtz  1/1    Running  0      13s
aras-demo-app-5b58b48d6d-spwkm  1/1    Running  0      8m29s
aras-demo-app-5b58b48d6d-zrtlx  1/1    Running  0      13s
```

Screenshot W6-3: KEDA scale-up — 4 replicas Running.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:53:17 PM CEST
in 0.210s (0)
NAME          SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  A
GE
aras-demo-keda  apps/v1.Deployment  aras-demo-app    1    8    True   False   False     False   prometheus                2
2m
---
aras-demo-app-5b58b48d6d-2q7pp  1/1    Running  0      11s
aras-demo-app-5b58b48d6d-82m24  1/1    Running  0      11s
aras-demo-app-5b58b48d6d-99s8c  1/1    Running  0      11s
aras-demo-app-5b58b48d6d-dhfhb  1/1    Running  0      26s
aras-demo-app-5b58b48d6d-ftqmw  1/1    Running  0      11s
aras-demo-app-5b58b48d6d-q9qtz  1/1    Running  0      26s
aras-demo-app-5b58b48d6d-spwkm  1/1    Running  0      8m42s
aras-demo-app-5b58b48d6d-zrtlx  1/1    Running  0      26s
```

Screenshot W6-4: KEDA scale-up peak — 8 replicas Running (maxReplicaCount reached).

```

80% 19
90% 34
95% 43
98% 53
99% 61
100% 147 (longest request)
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ ssh buth11@192.168.122.10 "ab -n 200000 -c 50 http://localhost:8080/"
This is ApacheBench, Version 2.3 <$Revision: 1903618 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking localhost (be patient)
Completed 20000 requests
Completed 40000 requests
Completed 60000 requests
Completed 80000 requests
Completed 100000 requests
Completed 120000 requests
Completed 140000 requests
Completed 160000 requests
Completed 180000 requests
Completed 200000 requests
Finished 200000 requests

Server Software:
Server Hostname: localhost
Server Port: 8080

Document Path: /
Document Length: 238 bytes

Concurrency Level: 50
Time taken for tests: 72.781 seconds
Complete requests: 200000
Failed requests: 121082
  (Connect: 0, Receive: 0, Length: 121082, Exceptions: 0)
Total transferred: 71226416 bytes
HTML transferred: 47626416 bytes
Requests per second: 2747.95 [#/sec] (mean)
Time per request: 18.195 [ms] (mean)
Time per request: 0.364 [ms] (mean, across all concurrent requests)
Transfer rate: 955.70 [Kbytes/sec] received

Connection Times (ms)
              min      mean[+/-sd] median   max
Connect:        0       2   1.4      1       9
Processing:      0      16  29.0      4     495
Waiting:        0      15  28.4      3     494
Total:          0      18  29.0      7     496

Percentage of the requests served within a certain time (ms)
 50%    7
 66%   11
 75%   20
 80%   26
 90%   48
 95%   87
 98%  100
 99%  105
100%  496 (longest request)
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$

```

Screenshot W6-5: ab 200000 requests — 2747 req/s, 72.781s.

```

Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:59:49 PM CEST
in 0.163s (0)
NAME          SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  A
GE
aras-demo-keda  apps/v1.Deployment  aras-demo-app    1    8    True   False   False     False   prometheus                2
9m
---
aras-demo-app-5b58b48d6d-spwkm  1/1    Running    0      15m

```

Screenshot W6-6: KEDA scale-down — ACTIVE: False, 1 replica. 30s cooldown elapsed.

Chapter 22: Results and Conclusions (Parts 1-6)

Metric	Part 1	Part 2	Part 3	Part 4	Part 5	Part 6
Nodes	3/3 Ready	—	—	—	—	—
SLO	100%	—	—	—	—	—
Cost	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
CI/CD	—	DevOps 4m24s	—	—	—	—
Ingress TLS	—	—	Manual	—	cert-manager	—
Autoscaling	—	—	HPA 2-8	—	—	KEDA 1-8
Scale to zero	—	—	No	—	—	Yes
Cooldown	—	—	5 min	—	—	30 sec
GitOps	—	—	—	FluxCD	—	—
CVE scan	—	—	—	1 CRIT+12 HIGH	—	—
Runtime sec.	—	—	—	Defender	—	—
Auto TLS	—	—	—	—	cert-manager	—
Issues	6	8	10	11	12	13

Chapter 23: Troubleshooting Log (13 issues)

ISSUE-001 — KQL: CPUCapacityNanoCores column missing

Resolution: Use getschema to verify available columns

ISSUE-002 — Azure Policy: missing cpuLimit/memoryLimit params

Resolution: Add --params to az policy assignment create

ISSUE-003 — Terraform: Resource Group already exists

Resolution: terraform import azurerm_resource_group.sre_lab

ISSUE-004 — Terraform: stale plan after import

Resolution: Generate new terraform plan -out=tfplan

ISSUE-005 — Container Insights missing after VM restart

Resolution: az k8s-extension create --name azuremonitor-containers

ISSUE-006 — POST-001: etcd Split-Brain (ks3 vs k3s)

Resolution: kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent

ISSUE-007 — Azure DevOps: TerraformInstaller task missing

Resolution: Install from Marketplace: ms-devlabs.custom-terraform-tasks

ISSUE-008 — Pipeline: permission for Variable Group

Resolution: Click View → Permit — one-time authorization

ISSUE-009 — sudo does not expand ~ in paths

Resolution: Use full path /home/buth11/ instead of ~/

ISSUE-010 — HPA shows cpu: unknown

Resolution: Add resources.requests to Deployment: cpu: 10m

ISSUE-011 — RBAC RoleBinding: apiRef instead of apiGroup

Resolution: Correct field: apiGroup: rbac.authorization.k8s.io

ISSUE-012 — cert-manager webhook: duration < 1h rejected

Resolution: Minimum duration: 1h5m, renewBefore: 6m

ISSUE-013 — KEDA: conflict with existing HPA

Resolution: Delete HPA before deploying ScaledObject

PART 7

Helm — Custom Chart aras-demo-app

Custom Helm chart v0.1.0 for aras-demo-app. Templates: Deployment, Service, Ingress (cert-manager), RBAC, NetworkPolicy, NOTES.txt.

Demonstrated: helm install, helm upgrade --set, helm rollback, helm history.

=== HELM ===

NAME	NAMESPACE	REVISION	UPDATED	STATUS	CHAR	
T	APP VERSION					
aras-demo	default	3	2026-06-06 17:02:26.737603427 +0200 CEST	deployed	aras	
-demo-app-0.1.0	1.0.0					
REVISION	UPDATED		STATUS	CHART	APP VERSION	DESCRIPTION
1	Sat Jun 6 17:00:57 2026		superseded	aras-demo-app-0.1.0	1.0.0	Install comp
lete						
2	Sat Jun 6 17:01:56 2026		superseded	aras-demo-app-0.1.0	1.0.0	Upgrade comp
lete						
3	Sat Jun 6 17:02:26 2026		deployed	aras-demo-app-0.1.0	1.0.0	Rollback to
1						

Screenshot W7-1: helm list — aras-demo deployed REVISION 3 (after rollback). helm history — REVISION 1 Install, REVISION 2 Upgrade (3 replicas), REVISION 3 Rollback to 1.

```
=== PODS ===
```

aras-demo-6ff6ffd5b-dqrgz	2/2	Running	0	27m
aras-demo-6ff6ffd5b-fq6fg	2/2	Running	0	27m
aras-demo-app-5b58b48d6d-spwkm	1/1	Running	1 (90m ago)	10d

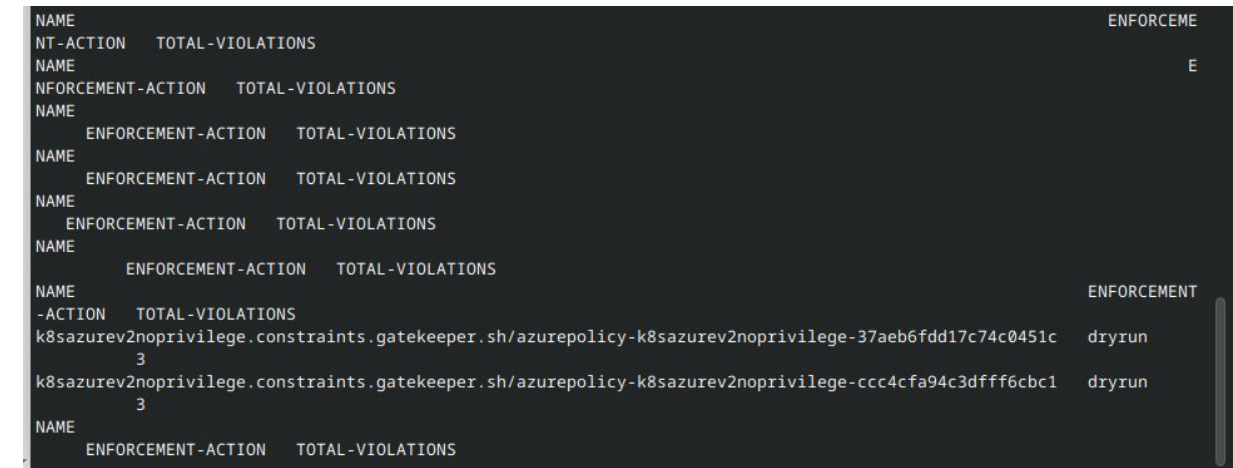
Screenshot W7-2: kubectl get pods — 2x aras-demo 2/2 Running (Linkerd sidecar), 1x old pod 1/1.

PART 8

OPA/Gatekeeper — Policy as Code Audit

Gatekeeper managed by Azure Arc — 16 ConstraintTemplates from Azure Policy. Mode: dryrun — audit violations without blocking. Custom ConstraintTemplates blocked by Azure webhook.

Policy	Violations	Mode
k8sazurev3containerlimits	34	dryrun
k8sazurev2containerallowedimages	30	dryrun
k8sazurev2blockautomounttoken	21	dryrun
k8sazurev3allowedusersgroups	22	dryrun
k8sazurev2noprivilege	0	dryrun
k8sazurev3disallowedcapabilities	0	dryrun



Screenshot W8-1: `kubectl get constraints` — `k8sazurev2noprivilege`: dryrun, 0 violations (application compliant with no-privilege policy). Full list of 16 ConstraintTemplates from Azure Policy.

PART 9

Chaos Engineering — Chaos Mesh

Chaos Mesh installed with K3s containerd configuration. PodChaos pod-kill experiment every 30 seconds. Self-healing verified — new pods appear within 3-5 seconds.

```
=== CHAOS MESH ===
NAME                                READY   STATUS    RESTARTS   AGE
chaos-controller-manager-85cbb588b9-25hcn 1/1     Running   0           67m
chaos-controller-manager-85cbb588b9-7wr8b 1/1     Running   0           67m
chaos-controller-manager-85cbb588b9-1mj4z 1/1     Running   0           67m
chaos-daemon-btxgv                     1/1     Running   0           67m
chaos-daemon-rprt6                     1/1     Running   0           67m
chaos-daemon-wd9cx                     1/1     Running   0           67m
chaos-dashboard-5b77585c6c-2ngtg        1/1     Running   0           67m
chaos-dns-server-6bccb5bb76-g7xtw       1/1     Running   0           67m
=== LINKERD ===
NAME      MESHED   SUCCESS   RPS   LATENCY P50   LATENCY P95   LATENCY P99   TCP_CONN
```

Screenshot W9-1: `kubectl get pods -n chaos-mesh` — 8 pods Running: 3x chaos-controller-manager, 3x chaos-daemon (DaemonSet — one per node), chaos-dashboard, chaos-dns-server.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | ... fedora: Wed 27 May 2026 12:53:17 PM CEST
in 0.210s (0)
NAME      SCALETARGETKIND  SCALETARGETNAME  MIN  MAX  READY  ACTIVE  FALLBACK  PAUSED  TRIGGERS  AUTHENTICATIONS  A
aras-demo-keda  apps/v1.Deployment  aras-demo-app    1    8    True   False   False     False   prometheus                2
---
aras-demo-app-5b58b48d6d-2q7pp 1/1    Running   0      11s
aras-demo-app-5b58b48d6d-82m24 1/1    Running   0      11s
aras-demo-app-5b58b48d6d-99s8c 1/1    Running   0      11s
aras-demo-app-5b58b48d6d-dhfhb 1/1    Running   0      26s
aras-demo-app-5b58b48d6d-ftqmw 1/1    Running   0      11s
aras-demo-app-5b58b48d6d-q9qtz 1/1    Running   0      26s
aras-demo-app-5b58b48d6d-spwkm 1/1    Running   0      8m42s
aras-demo-app-5b58b48d6d-zrtlx 1/1    Running   0      26s
```

Screenshot W9-2: During PodChaos experiment — 8 aras-demo pods Running simultaneously. Kubernetes immediately replaced killed pods with new ones.

PART 10

Linkerd Service Mesh — mTLS

Linkerd edge-26.6.1 installed with viz extension. Namespace default annotated — sidecar proxy automatically injected into all new pods. MESHED: 2/2.

```
aras-demo      2/2      -      -      -      -      -      -  
=== OPA ===  
NAME                                                    ENFORCEME  
NT-ACTION  TOTAL-VIOLATIONS
```

Screenshot W10-1: linkerd viz stat deployment/aras-demo — MESHED: 2/2. Both pods in service mesh with active mTLS.

```
=== PODS ===  
aras-demo-6ff6ffd5b-dqrgz      2/2      Running  0          27m  
aras-demo-6ff6ffd5b-fq6fg      2/2      Running  0          27m  
aras-demo-app-5b58b48d6d-spwkm 1/1      Running  1 (90m ago) 10d
```

Screenshot W10-2: kubectl get pods -n default — aras-demo 2/2 Running (2 containers: whoami app + linkerd-proxy sidecar). Old aras-demo-app pod 1/1 without sidecar.

Results Summary — Parts 1-10

Metric	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Cost	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
SLO	100%	—	—	—	—	—	—	—	—	—
CI/CD	—	4m24s	—	—	—	—	—	—	—	—
TLS	—	—	Manual	—	Auto	—	—	—	—	—
Autoscaling	—	—	HPA	—	—	KEDA	—	—	—	—
GitOps	—	—	—	FluxCD	—	—	—	—	—	—
CVE scan	—	—	—	1 CRIT	—	—	—	—	—	—
Helm	—	—	—	—	—	—	v0.1.0	—	—	—
Policy	—	—	—	—	—	—	—	34 viol.	—	—
Chaos	—	—	—	—	—	—	—	—	pod-kill	—
mTLS	—	—	—	—	—	—	—	—	—	2/2
Issues	6	8	10	11	12	13	14	15	16	17

Troubleshooting Log — 17 issues

ISSUE-001 — KQL: CPUCapacityNanoCores missing

Resolution: Use getschema to verify columns

ISSUE-002 — Azure Policy: missing params

Resolution: Add --params

ISSUE-003 — Terraform: RG already exists

Resolution: terraform import

ISSUE-004 — Terraform: stale plan

Resolution: Regenerate after import

ISSUE-005 — Container Insights after restart

Resolution: Reinstall az k8s-extension

ISSUE-006 — etcd Split-Brain ks3 vs k3s

Resolution: kubectrl delete node + re-register

ISSUE-007 — Azure DevOps: TerraformInstaller

Resolution: Install from Marketplace

ISSUE-008 — Pipeline: Variable Group perm

Resolution: View → Permit

ISSUE-009 — sudo ~ expansion

Resolution: Use full path /home/buth11/

ISSUE-010 — HPA cpu: unknown

Resolution: Add resources.requests

ISSUE-011 — RBAC: apiRef vs apiGroup

Resolution: Fix to apiGroup: rbac.authorization.k8s.io

ISSUE-012 — cert-manager: duration < 1h

Resolution: Min duration 1h5m, renewBefore 6m

ISSUE-013 — KEDA: conflict with HPA

Resolution: Delete HPA first

ISSUE-014 — Helm: SA already exists

Resolution: Delete old SA before install

ISSUE-015 — OPA: custom template blocked

Resolution: Azure Arc manages Gatekeeper

ISSUE-016 — Chaos Mesh: scheduler removed

Resolution: Use duration instead

ISSUE-017 — Linkerd: no sh in container

Resolution: Use linkerd-proxy container